

Goodness of Estimators

Learning Outcomes

- Consistency
- Sufficiency
- Information
- Efficiency

Consistency

An estimator is considered a consistent estimator of θ if the estimator, on average, converges to θ as $n \rightarrow \infty$.

Consistency

Let $X_1, \dots, X_n$ be a random sample from a distribution with parameter θ . The estimator $\hat{\theta}$ is a consistent estimator of the θ if

1. $E\{(\hat{\theta} - \theta)^2\} \rightarrow 0$ as $n \rightarrow \infty$
2. $P(|\hat{\theta} - \theta| \geq \epsilon) \rightarrow 0$ as $n \rightarrow \infty$ for every $\epsilon > 0$

Sufficiency

Sufficiency evaluates whether a statistic (or estimator) contains enough information of a parameter θ . In essence a statistic is considered sufficient to infer θ if it provides enough information about θ .

Sufficiency

Let $X_1, \dots, X_n$ be a random sample from a distribution with parameter θ . A statistic $T = t(X_1, \dots, X_n)$ is said to be sufficient for making inferences of a parameter θ if condition joint distribution of $X_1, \dots, X_n$ given $T = t$ does not depend on θ .

Factorization Theorem

The **Factorization Theorem** provides a condition for a statistic $T(X)$ to be sufficient for a parameter θ given a probability density function or probability mass function.

Factorization Theorem

Let $X = (X_1, X_2, \dots, X_n)$ be a random sample with joint probability density (or mass) function $f(x|\theta)$, where θ is a parameter.

Factorization Theorem

Theorem: A statistic $T(X)$ is sufficient for θ if and only if the joint density (or mass) function $f(x|\theta)$ can be factored into the form

$$f(x|\theta) = g(T(x), \theta) \cdot h(x)$$

where:

- $g(T(x), \theta)$ is a function that depends on $T(x)$ and θ ,
- $h(x)$ is a function that does not depend on θ .

Factorization Theorem

In other words, $f(x|\theta)$ can be written as a product of two functions, where only one function depends on the parameter θ and the sufficient statistic $T(X)$.

Implications: The Factorization Theorem is useful for identifying sufficient statistics, which summarize all necessary information from a sample about the parameter θ .

Information

In Statistics, information is thought of as how much does the data tell you about a parameter θ . In general, the more data is provided, the more information is provided to estimate θ .

Information

Information can be quantified using Fisher's Information $I(\theta)$. For a single observation, Fisher's Information is defined as

$$I(\theta) = E \left[-\frac{\partial^2}{\partial \theta^2} \log\{f(X; \theta)\} \right],$$

where $f(X; \theta)$ is either the PMF or PDF of the random variable X .

Furthermore, $I(\theta)$ can be defined as

$$I(\theta) = \text{Var} \left\{ \frac{\partial}{\partial \theta} \log f(X; \theta) \right\}.$$

Show the following property:

$$E \left[-\frac{\partial^2}{\partial \theta^2} \log \{ f(X; \theta) \} \right] = \text{Var} \left\{ \frac{\partial}{\partial \theta} \log f(X; \theta) \right\}$$

Efficiency

Efficiency of an estimator T is the ratio of variation compared to the lowest possible variance.

Efficiency

The efficiency of an estimator T , where T is an unbiased estimator of θ , is defined as

$$\text{efficiency of } T = \frac{1}{\text{Var}(T)nI(\theta)}$$